

Original practice 3: Heating, power and energy budgets

Original prediction practice—not official questions or a guarantee

Total: 36 marks • Suggested time: 75 minutes

Joule heating, dimensional checks, fixed-current versus fixed-voltage comparisons and energy-use evidence.

Status and instructions

Original prediction practice—not official questions or a guarantee

Author-created chapter practice, not a claim about an official examination format. All 16 questions are compulsory. Show working and units. Circuit wires, meters and sources are ideal unless stated; resistor temperatures remain constant unless a question explicitly studies a change. Treat stated data as exact unless a tolerance is given.

All case datasets and SVGs are original constructed teaching material, not measurements reported by NCERT. NCERT is cited for concepts and formulas, not as the source of these questions.

- A: Q1-6, $6 \times 1 = 6$ marks
- B: Q7-10, $4 \times 2 = 8$ marks
- C: Q11-13, $3 \times 3 = 9$ marks
- D: Q14, $1 \times 5 = 5$ marks
- E: Q15-16, $2 \times 4 = 8$ marks
- Total: 16 questions, 36 marks

Official curriculum scope and assessment limitation

CBSE Science (086), Class X, 2026-27, PDF p.6 Unit IV lists electric current, potential difference, resistance, resistivity, Ohm's law, resistor combinations, heating and power. Practical 4 on PDF p.7 covers V-I measurements; practical 5 on PDF p.8 covers series and parallel equivalent resistance. Source: https://cbseacademic.nic.in/web_material/CurriculumMain27/SecPart1/Science_SecP1_2026-27.pdf

The same PDF p.6 contains an unresolved teacher-note wording inconsistency naming "Electric Effects of Electric Current" among unassessed topics, despite the explicit Unit IV list. These are chapter study materials, not a guarantee of summative assessment coverage; consult school/CBSE clarification. Unit IV's 13 marks covers a combined unit, not an Electricity-only allotment.

Section A — MCQ

1. [1 marks]

1. A 6 V resistor draws 0.40 A. What is its electrical power?

- A. 15 W
- B. 0.067 W
- C. 2.4 W
- D. 24 W

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2. [1 marks]

2. For the same resistor and heating time, the current becomes three times as large. The heat becomes:

- A. Nine times
- B. Three times
- C. One third
- D. Unchanged

Source/editorial note: Original prediction practice—not official questions or a guarantee. Original question and worked answer. Concept/formula references: §11.7, printed pp188-190; Eqs. (11.19)–(11.21) pp188-189: $P = VI$, $H = VIt = I^2Rt$. Heating-element and filament applications: §11.7.1 p190; alloy properties §11.5 p179. Official chapter: <https://ncert.nic.in/textbook/pdf/jesc111.pdf>

3. [1 marks]

3. Which amount of energy equals 0.25 kWh?

- A. 250 J
- B. 900 J
- C. 9.0×10^3 J
- D. 9.0×10^5 J

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4. [1 marks]

4. At the same fixed voltage, a resistor's resistance is doubled while remaining ohmic. Its power becomes:

- A. Twice as large
- B. Half as large
- C. Four times as large
- D. Unchanged

Source/editorial note: Original prediction practice—not official questions or a guarantee. Original question and worked answer. Concept/formula references: §11.8, printed pp191-192; Eqs. (11.22)–(11.23) p191: $P = VI = I^2R = V^2/R$; $E = Pt$; 1 kWh = 3.6×10^6 J. Official chapter: <https://ncert.nic.in/textbook/pdf/jesc111.pdf>

5. [1 marks]

5. A label states 12 V, 18 W for a resistive device at its operating temperature. What is its resistance at that temperature?

- A. 1.5 Ω
- B. 216 Ω
- C. 8 Ω
- D. 30 Ω

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6. [1 marks]

6. Which statement correctly distinguishes power and energy?

- A. Watt measures power; kilowatt hour measures energy
- B. Both are measured in ampere
- C. Kilowatt hour measures power
- D. Watt measures total charge

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Section B — 2-mark questions

7. [2 marks]

7. A 0.80 A current flows through a 5 Ω resistor for 90 s. Calculate the heat and average power.

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8. [2 marks]

8. A 24 W device works for 25 minutes. Express the energy used in (a) joules and (b) Wh.

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9. [2 marks]

9. Choose a suitable general class of material for a compact heating coil rather than a low-resistivity connecting lead, and explain two relevant properties.

Source/editorial note: Original prediction practice—not official questions or a guarantee. Original question and worked answer. Concept/formula references: §11.5, printed pp177–180; Activity 11.3 pp177–178; Eqs. (11.8)–(11.10) p178: $R = \rho l/A$; material and temperature discussion p179. §11.7, printed pp188–190; Eqs. (11.19)–(11.21) pp188–189: $P = VI$, $H = VIt = I^2Rt$. Heating-element and filament applications: §11.7.1 p190; alloy properties §11.5 p179. Official chapter: <https://ncert.nic.in/textbook/pdf/jesc111.pdf>

10. [2 marks]

10. A $12\ \Omega$ resistor and a $6\ \Omega$ resistor carry the same 0.50 A for the same time. Which produces more heat, and by what factor? Explain why this is not the fixed-voltage result.

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Section C — 3-mark questions

11. [3 marks]

11. A 9 V source sends 180 C through a resistive load in 2 minutes. Calculate (a) current, (b) energy transferred and (c) power.

Source/editorial note: Original prediction practice—not official questions or a guarantee. Original question and worked answer. Concept/formula references: §11.1, printed pp171–172, Eq. (11.1): $I = Q/t$; conventional current opposes electron flow. §11.2, printed pp173–174, Eqs. (11.2)–(11.3): $V = W/Q$; voltmeter in parallel. §11.8, printed pp191–192; Eqs. (11.22)–(11.23) p191: $P = VI = I^2R = V^2/R$; $E = Pt$; $1\text{ kWh} = 3.6 \times 10^6\text{ J}$. Official chapter: <https://ncert.nic.in/textbook/pdf/jesc111.pdf>

12. [3 marks]

12. A 40 W display runs for 3 h; a 150 W heating pad runs for 24 min. Calculate each energy in Wh and identify which uses more, with the difference.

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13. [3 marks]

13. A resistive cartridge is labelled 20 V , 50 W . Assuming its resistance stays constant, calculate (a) rated resistance, (b) current at 10 V and (c) power at 10 V .

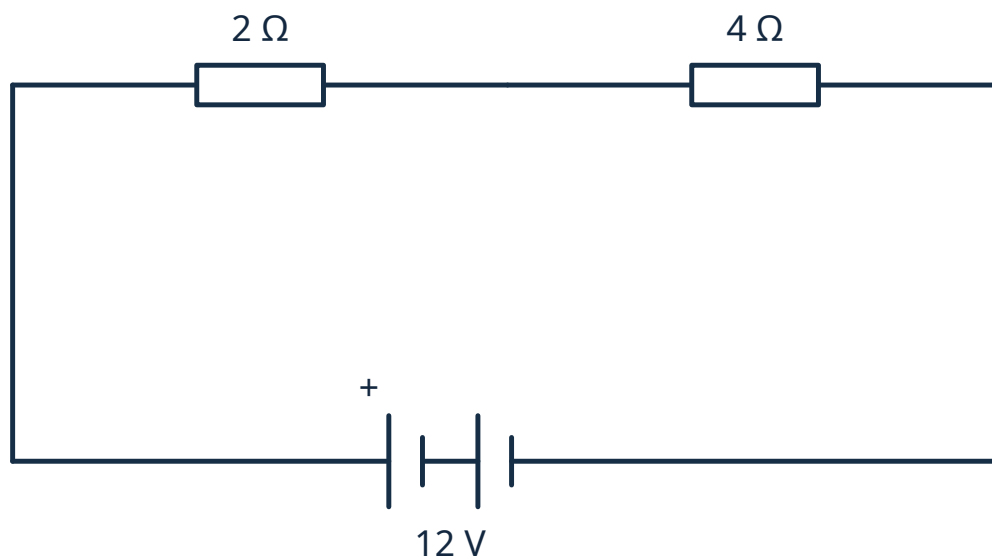
Source/editorial note: Original prediction practice—not official questions or a guarantee. Original question and worked answer. Concept/formula references: §11.8, printed pp191–192; Eqs. (11.22)–(11.23) p191: $P = VI = I^2R = V^2/R$; $E = Pt$; $1\text{ kWh} = 3.6 \times 10^6\text{ J}$. §11.4, printed pp175–177; Activity 11.1 p175; Eqs. (11.4)–(11.7) p176: $V = IR$ at constant temperature; Activity 11.2 pp176–177 compares components. Official chapter: <https://ncert.nic.in/textbook/pdf/jesc111.pdf>

Section D — 5-mark questions

14. [5 marks]

14. The supplied 12 V circuit contains $2\ \Omega$ and $4\ \Omega$ heating resistors in series. (a) Find current (1). (b) Find the power in each resistor (2). (c) Find total heat in 2 minutes (1). (d) Check this heat against source energy using VIt (1).

12 V series heating circuit



Source/editorial note: Original prediction practice—not official questions or a guarantee. Original question and worked answer. Concept/formula references: §11.6.1, printed pp182–185; Activity 11.4 p182 (same current); Activity 11.5 p183 (voltage sum); Eqs. (11.11)–(11.14) pp183–184. §11.7, printed pp188–190; Eqs. (11.19)–(11.21) pp188–189: $P = VI$, $H = VIt = I^2Rt$. Heating-element and filament applications: §11.7.1 p190; alloy properties §11.5 p179. §11.8, printed pp191–192; Eqs. (11.22)–(11.23) p191: $P = VI = I^2R = V^2/R$; $E = Pt$; $1\ \text{kWh} = 3.6 \times 10^6\ \text{J}$. Official chapter: <https://ncert.nic.in/textbook/pdf/jesc111.pdf>

Section E — Case-based questions

15. [4 marks]

15. Constructed energy audit: a club uses a 30 W lamp for 4 h/day and a 60 W soldering station for 0.5 h/day, for 20 days. Treat powers as constant; energy costs ₹8/kWh with no other charges. (a) Find lamp energy over 20 days (1). (b) Find station energy (1). (c) Find total cost (1). (d) Find the energy saved by cutting lamp use by 1 h/day over those 20 days (1).

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16. [4 marks]

16. Constructed controlled-heating dataset: one resistor gives trial A: $I = 0.50\text{ A}$, $t = 40\text{ s}$, $H = 30\text{ J}$; trial B: $I = 1.00\text{ A}$, $t = 40\text{ s}$, $H = 120\text{ J}$; trial C: $I = 0.50\text{ A}$, $t = 80\text{ s}$, $H = 60\text{ J}$. H denotes electrical energy dissipated, not a thermometer estimate. (a) Infer R from A (1). (b) Explain B/A (1). (c) Explain C/A (1). (d) Predict H at 0.75 A for 40 s with the same R (1).

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Solutions and verification status

Only explicitly verified solutions are printed. A missing solution is not a claim that the question is unsolvable.

Question 1 — original-practice-03-q01

[1 mark] C. $P = VI = 6 \times 0.40 = 2.4 \text{ W}$.

Question 2 — original-practice-03-q02

[1 mark] A. $H = I^2Rt$, so heat is multiplied by $3^2 = 9$.

Question 3 — original-practice-03-q03

[1 mark] D. $0.25 \times 3.6 \times 10^6 = 9.0 \times 10^5 \text{ J}$.

Question 4 — original-practice-03-q04

[1 mark] B. $P = V^2/R$, so doubling R halves P at fixed V .

Question 5 — original-practice-03-q05

[1 mark] C. $R = V^2/P = 144/18 = 8 \Omega$.

Question 6 — original-practice-03-q06

[1 mark] A. Power is the rate of energy transfer; energy = power $\times$ time for constant power.

Question 7 — original-practice-03-q07

[1 mark] $H = I^2Rt = 0.80^2 \times 5 \times 90 = 288 \text{ J}$.

[1 mark] $P = H/t = 288/90 = 3.2 \text{ W}$ (also I^2R).

Question 8 — original-practice-03-q08

[1 mark] (a) $t = 1500 \text{ s}$; $E = Pt = 24 \times 1500 = 36\,000 \text{ J}$.

[1 mark] (b) $t = 25/60 \text{ h}$; $E = 24 \times 25/60 = 10 \text{ Wh}$.

Question 9 — original-practice-03-q09

[1 mark] Use a suitable alloy such as nichrome: its relatively high resistivity permits appreciable resistance in a compact coil.

[1 mark] Suitable heating alloys resist oxidation at high temperature, improving survival during heating.

Question 10 — original-practice-03-q10

[1 mark] At equal I and t , $H \propto R$, so 12Ω produces twice as much heat as 6Ω .

[1 mark] At equal V and t , $H = V^2t/R$ instead; the lower resistance would produce more heat. The held-constant variable differs.

Question 11 — original-practice-03-q11

[1 mark] (a) $t = 120 \text{ s}$; $I = Q/t = 180/120 = 1.5 \text{ A}$.

[1 mark] (b) $E = VQ = 9 \times 180 = 1620 \text{ J}$.

[1 mark] (c) $P = E/t = 1620/120 = 13.5 \text{ W}$.

Question 12 — original-practice-03-q12

[1 mark] Display: $E = 40 \times 3 = 120 \text{ Wh}$.

[1 mark] Pad: $E = 150 \times (24/60) = 60 \text{ Wh}$.

[1 mark] The display uses more by $120 - 60 = 60 \text{ Wh}$; a larger power does not alone imply larger energy.

Question 13 — original-practice-03-q13

[1 mark] (a) $R = 20^2/50 = 8 \Omega$.

[1 mark] (b) $I = 10/8 = 1.25 \text{ A}$.

[1 mark] (c) $P = 10^2/8 = 12.5 \text{ W}$. The fixed-resistance assumption is essential.

Question 14 — original-practice-03-q14

[1 mark] (a) $R_{\text{total}} = 2 + 4 = 6 \Omega$; $I = 12/6 = 2 \text{ A}$.

[1 mark] (b) $P_2 = I^2 \times 2 = 4 \times 2 = 8 \text{ W}$.

[1 mark] (b) $P_4 = I^2 \times 4 = 4 \times 4 = 16 \text{ W}$.

[1 mark] (c) $t = 120 \text{ s}$; $H_{\text{total}} = (8 + 16) \times 120 = 2880 \text{ J}$.

[1 mark] (d) $VIt = 12 \times 2 \times 120 = 2880 \text{ J}$, agreeing with the summed resistor heating.

Question 15 — original-practice-03-q15

[1 mark] (a) $E_{\text{lamp}} = 0.030 \text{ kW} \times 4 \text{ h/day} \times 20 \text{ days} = 2.4 \text{ kWh}$.

[1 mark] (b) $E_{\text{station}} = 0.060 \times 0.5 \times 20 = 0.6 \text{ kWh}$.

[1 mark] (c) Total = 3.0 kWh; cost = $3.0 \times ₹8 = ₹24$.

[1 mark] (d) Saving = $0.030 \times 1 \times 20 = 0.6 \text{ kWh}$.

Question 16 — original-practice-03-q16

[1 mark] (a) $R = H/(I^2t) = 30/(0.50^2 \times 40) = 3 \Omega$.

[1 mark] (b) Doubling current at fixed R and t multiplies heat by four: $120/30 = 4$.

[1 mark] (c) Doubling time at fixed I and R doubles heat: $60/30 = 2$.

[1 mark] (d) $H = 0.75^2 \times 3 \times 40 = 67.5 \text{ J}$.